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Yndestad, S., Austreid, E., Knappskog, S., Chrisanthar, R., Lilleng, P. K., Lønning, P. E., & Eikesdal, H. P. (2017). High PTEN gene expression is a negative prognostic marker in human primary breast cancers with preserved p53 function. *Breast Cancer Research and Treatment*, 163, 177-190.

유방암 환자 282명을 대상으로 한 분석에서 PTEN mRNA 발현이 높은 그룹은 낮은 그룹 대비 재발 위험이 유의하게 높았다.

HR = 1.78 > 1, 즉 cox model에서 exponential term이 1보다 큰 것으로 보아, $\beta > 0$ 으로 해석할 수 있다. PTEN은 종양억제유전자임에도 불구하고 mRNA 발현이 높을 때, prognosis가 좋지 않다는 결과를 시사한다.

(p.183)

Prognostic impact of *PTEN* gene expression

Excluding patients with stage 4 disease from the analysis, high *PTEN* gene expression, defined as a *PTEN* mRNA level above the median, was associated with significantly shorter RFS (hazard ratio (HR) for recurrence 1.78, 95% confidence interval (CI) 1.26–2.50, $p = 0.001$), and DSS (HR for breast cancer-specific death 1.72, 95% CI 1.20–2.47, $p = 0.003$) across the pooled cohort of patients with stage 3 disease ($n = 282$, Fig. 3a–d). Among tumors wild-type for *TP53*, a high *PTEN* level remained a negative prognostic marker, with inferior RFS as well as DSS (HR 1.82, 95% CI 1.22–2.72, $p = 0.003$ and HR 1.78, 95% CI 1.16–2.73, $p = 0.009$, respectively; Figs. 3c, d, 4a, b). In contrast, no significant association between outcome and *PTEN* gene expression level was observed in patients with tumors harboring *TP53* mutations (Fig. 3c, d, 4c, d). These findings were consistent across each individual trial (Online Resource 3).

2.

Cizkova, M., Vacher, S., Meseure, D., Trassard, M., Susini, A., Mlcuchova, D., Callens, C., Rouleau, E., Spyratos, F., Lidereau, R., & Bièche, I. (2013). PIK3R1 underexpression is an independent prognostic marker in breast cancer. *BMC Cancer*, 13, 545.

유방암 환자 458명을 대상으로 한 분석에서 PIK3R1 mRNA 발현이 정상인 그룹은 저발현 그룹 대비 전이 위험이 유의하게 낮았으며(HR = 0.59), 이는 Cox model에서 $\beta < 0$ 에 해당한다. 즉, PIK3R1 mRNA 발현이 높을 때, prognosis가 좋다는 결과를 시사한다.

only *PIK3CA* mutations and *PIK3R1* underexpression, as separate markers, were associated with MFS and had opposite effects on patient survival: *PIK3CA* mutation was associated with better MFS and *PIK3R1* underexpression was associated with poorer MFS ($p = 0.016$ and $p = 0.00028$, respectively). *PIK3R1* underexpres-

(p.9~p.10)

Underexpression	1	1	1	1	1	1	1	1				
Normal	0.59 (0.43-0.81)	0.0013	0.62 (0.44-0.88)	0.0076	NS	0.65 (0.43-0.99)	0.043	NS	0.40 (0.19-0.85)	0.018	0.66 (0.46-0.95)	0.024

결론

같은 PI3K-AKT pathway 내에서 PTEN($\beta > 0$)과 PIK3R1($\beta < 0$)이 반대 부호를 가짐으로써 β 부호 혼합 설계가 정당하다.